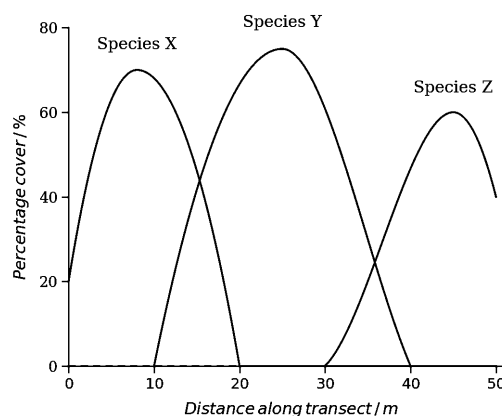


- 1 An ecological investigation was carried out to study the distribution and abundance of three plant species (X, Y, and Z) along a coastal sand dune.

A continuous 50 m belt transect was established from the edge of the beach inland, using contiguous 1 m × 1 m quadrats (50 quadrats in total). The percentage cover of each plant species was recorded in each quadrat. The results are shown in the graph below.

Which of the following statements is/are correct?

- 1 The frequency of occurrence of species Y across the entire 50 m transect is 60%.
- 2 At the 8 m position along the transect, the population density of species X is definitely higher than at the 18 m position.
- 3 At the 35 m position along the transect, species Y and species Z together cover the entire ground area within the quadrat.



- A none of them
- B 1 only
- C 2 only
- D 3 only
- E 1 and 2 only
- F 1 and 3 only
- G 2 and 3 only
- H 1, 2 and 3

2 Which of the following statements about the human circulatory system is/are correct?

1. The walls of capillaries consist of a single layer of cells, providing a short diffusion distance for substances.
2. The pulmonary artery transports oxygenated blood from the heart to the lungs.
3. During a cardiac cycle, the blood pressure in the aorta is significantly higher than the pressure in the pulmonary artery.

- A** none of them
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3

- 3 An investigation was carried out to study the effect of salinity on the germination of a crop species.

Three identical Petri dishes were prepared, each containing 20 viable seeds placed on filter paper moistened with a different concentration of sodium chloride solution: 0 mM NaCl (control), 50 mM NaCl, and 100 mM NaCl.

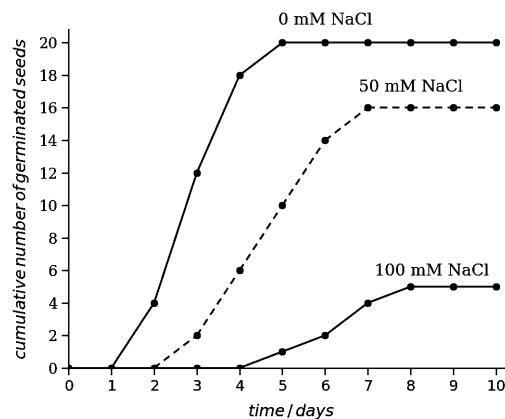
All other environmental variables, including temperature and light conditions, were kept constant.

The cumulative number of germinated seeds in each dish was recorded once every day for 10 days.

The results are shown in the graph.

Which of the following statements is/are correct?

- 1 Over the first 5 days, the average rate of germination was highest in the 0 mM NaCl solution.
- 2 At day 4, 30% of the seeds in the 50 mM NaCl solution remained ungerminated.
- 3 The reciprocal of the time taken for 50% of the seeds in a treatment to germinate can be used as a measure of the relative rate of germination.

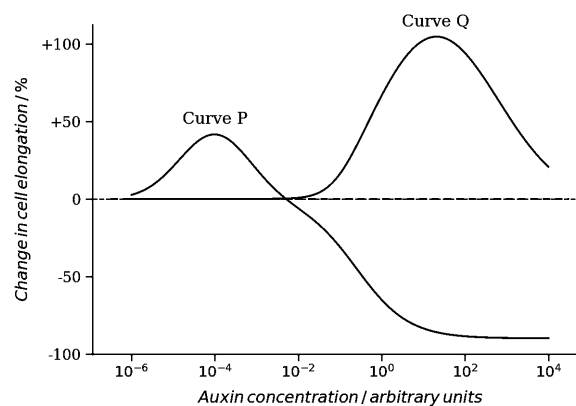


- A none of them
- B 1 only
- C 2 only
- D 3 only
- E 1 and 2 only
- F 1 and 3 only
- G 2 and 3 only
- H 1, 2 and 3

- 4 The graph shows the effect of different concentrations of auxin (IAA) on cell elongation in two different plant tissues, P and Q, measured as a percentage change relative to untreated controls (0%).

Which of the following statements is/are correct?

- 1 Curve P represents the response of root tissue, and Curve Q represents the response of shoot tissue.
- 2 In a horizontal seedling where auxin accumulates on the lower side to 1.0 arbitrary units while the upper side remains at 10^{-4} arbitrary units, differential elongation will cause the shoot to bend downwards.
- 3 In a vertical root with a uniform auxin concentration of 1.0 arbitrary units, applying an inhibitor that reduces the auxin concentration throughout the root to 10^{-4} arbitrary units will increase the root elongation rate.



- A 1 only
- B 2 only
- C 3 only
- D 1 and 2 only
- E 1 and 3 only
- F 2 and 3 only
- G 1, 2 and 3
- H none of the statements

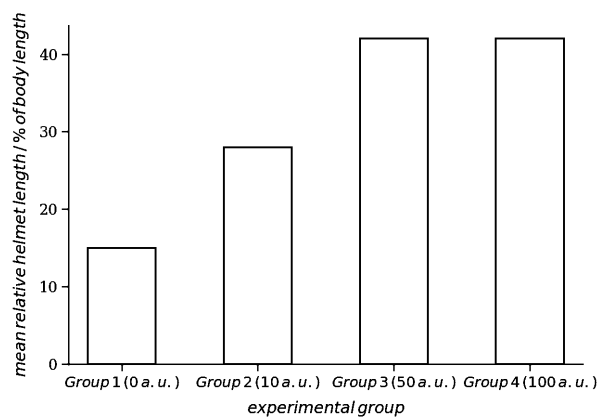
- 5 Genetically identical individuals of the water flea *Daphnia magna* were divided into four equal groups of twenty individuals. All four groups were raised in identical volumes of water, at the same temperature, with the same light cycle, and fed equal quantities of algae. The only difference between the groups was the concentration of a chemical cue (kairomone) released by a predator added to the water:

- Group 1: 0 a.u. of predator cue
- Group 2: 10 a.u. of predator cue
- Group 3: 50 a.u. of predator cue
- Group 4: 100 a.u. of predator cue

Upon reaching maturity, the mean relative helmet length (the length of the defensive helmet as a percentage of body length) was measured for each group. The results are shown in the graph.

Which of the following statements could explain these results?

- 1 The difference in mean relative helmet length between Group 1 and Group 2 is caused by genetic differences between the individuals in the two groups.
- 2 The identical mean relative helmet length in Group 3 and Group 4 could be due to an environmental factor, such as food availability, becoming limiting.
- 3 The identical mean relative helmet length in Group 3 and Group 4 could be due to reaching the maximum phenotypic limit determined by the genotype of the clone.



- A none of them
- B 1 only
- C 2 only
- D 3 only
- E 1 and 2 only
- F 1 and 3 only
- G 2 and 3 only
- H 1, 2 and 3

- 6 A short section of mRNA has the following sequence:

5' - AUG CUC CAC UAA - 3'

The genetic code for selected amino acids is provided in the table below:

Amino Acid	Codons
Histidine (His)	CAU, CAC
Glutamine (Gln)	CAA, CAG
Leucine (Leu)	CUU, CUC, CUA, CUG
Valine (Val)	GUU, GUC, GUA, GUG

A mutation results in the substitution of the cytosine (C) at nucleotide position 6 with a guanine (G).

Which of the following describes the effect on the polypeptide sequence?

- A** The amino acid sequence remains unchanged.
- B** Leu is replaced by Val.
- C** His is replaced by Gln.
- D** Leu is replaced by His.
- 7 Epidermolysis bullosa is an inherited condition in humans caused by a mutation in a gene that codes for a structural protein needed to anchor the outer layer of skin (epidermis) to underlying tissues.

In an experimental gene therapy treatment, skin stem cells were taken from an adult patient with this condition. A functional copy of the gene was inserted into the nuclear DNA of these stem cells using a viral vector. The genetically modified stem cells were then grown in the laboratory to form sheets of epidermal tissue, which were grafted onto the patient's skin to heal open wounds.

Which of the following statements about this gene therapy treatment is/are correct?

- 1 Transcription and translation of the inserted gene in the cells of the grafted epidermis can produce the functional structural protein.
- 2 When the genetically modified stem cells divide by mitosis, the inserted gene is replicated so that daughter cells retain the inserted gene.
- 3 The patient's biological children will inherit the inserted functional gene from this patient.
- A** 1 only
- B** 2 only
- C** 3 only
- D** 1 and 2 only
- E** 1 and 3 only
- F** 2 and 3 only
- G** 1, 2 and 3

- 8 Two cellular structures, structure X and structure Y, were imaged using electron microscopy. The micrographs are not shown.

Some of the data collected is shown in the table.

	structure X	structure Y
actual length / nm	2500	25
image length on micrograph / mm	50	1.5

Which of the following statements is/are correct?

- 1 Structure X has been magnified 20 000 times.
- 2 Structure X could be a mitochondrion and structure Y could be a ribosome.
- 3 Structure Y can be resolved using a standard light microscope.

- A** none of them
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3

- 9 A sealed flask containing a suspension of yeast (*Saccharomyces cerevisiae*) in a glucose nutrient solution was incubated at 25 °C. The rate of oxygen uptake, the rate of carbon dioxide release, and the rate of ethanol production were measured after 1 hour and after 6 hours.

	Time = 1 hour	Time = 6 hours
rate of O ₂ uptake / arbitrary units	8.4	0.2
rate of CO ₂ release / arbitrary units	8.5	5.6
rate of ethanol production / arbitrary units	0.1	5.5

Which of the following statements is/are correct?

- 1 A single molecule of glucose metabolised by the yeast yielded fewer molecules of ATP at 6 hours than at 1 hour.
- 2 At 6 hours, the oxidation of NADH to NAD⁺ was coupled to the reduction of oxygen at the inner mitochondrial membrane.
- 3 At 6 hours, some of the energy originally stored in glucose remained in an organic waste product released into the medium.

- A none of them
- B 1 only
- C 2 only
- D 3 only
- E 1 and 2 only
- F 1 and 3 only
- G 2 and 3 only
- H 1, 2 and 3

- 10** In a population of rats, resistance to the anticoagulant poison warfarin is determined by a single gene with two alleles, R and S .

The table provides information about the phenotypes associated with each genotype:

Genotype	Effect of standard dose of warfarin	Health in the absence of warfarin
SS	fatal internal bleeding	normal health and survival
RS	survives	normal health and survival
RR	survives	severe vitamin K deficiency (often fatal)

Which of the following statements is/are correct?

- 1 In habitats where warfarin is never applied, natural selection is likely to maintain a low frequency of the R allele in the population.
- 2 The application of warfarin causes the S allele to mutate into the R allele in surviving rats.
- 3 If all SS individuals in a population are killed by warfarin before they reproduce, the S allele will be completely eliminated from the next generation.

- A** none of them
B 1 only
C 2 only
D 3 only
E 1 and 2 only
F 1 and 3 only
G 2 and 3 only
H 1, 2 and 3

- 11 An agricultural investigation was carried out using two varieties of a crop plant, Variety X and Variety Y. For each variety, all plants used were genetically identical clones.

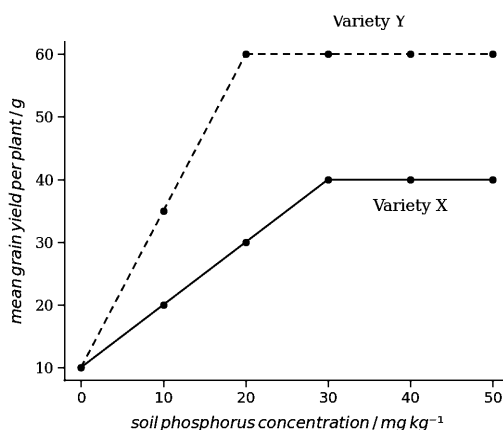
Groups of plants of each variety were grown across a range of soil phosphorus concentrations under strictly controlled environmental conditions:

- equal volumes of soil containing identical baseline nutrients
- identical temperature, light intensity, and photoperiod
- watered with equal volumes of water
- identical atmospheric CO₂ concentration

After 16 weeks of growth, the mean grain yield per plant was determined for each group. The results are shown in the graph below.

Which of the following statements is/are correct?

- 1 At a soil phosphorus concentration of 10 mg kg⁻¹, the difference in mean grain yield between Variety X and Variety Y is due to differences in their genotype.
- 2 For Variety X, the difference in mean grain yield between plants grown at 10 mg kg⁻¹ and plants grown at 25 mg kg⁻¹ is due to differences in their genotype.
- 3 Above 30 mg kg⁻¹, the plateau in mean grain yield for Variety X could be because another abiotic factor has become limiting.



- A none of them
B 1 only
C 2 only
D 3 only
E 1 and 2 only
F 1 and 3 only
G 2 and 3 only
H 1, 2 and 3

- 12** A botanist crosses a homozygous purple-flowered plant (genotype BB) with a homozygous white-flowered plant (genotype bb). Flower colour is controlled by a single autosomal gene, where the purple allele (B) is dominant to the white allele (b).

One of the offspring produced from this cross has entirely white flowers. Genetic analysis confirms that both parent plants are the true biological parents, and that all somatic cells and germ cells of this white-flowered offspring have the same genotype.

Which of the following statements could explain this result?

1 A loss-of-function mutation in the B allele occurred during meiosis in the BB parent.

2 A loss-of-function mutation in the B allele occurred during the first mitotic division of the embryo.

3 Chromosome non-disjunction during meiosis in the BB parent produced a gamete lacking the chromosome carrying the B gene.

- A** none of them
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3

- 13** The table shows some mRNA codons and the corresponding amino acids or translation stop signals:

Codon	Amino acid / function
AUG	Met
CCU	Pro
CCG	Pro
CGA	Arg
CGU	Arg
GAA	Glu
GAG	Glu
GAU	Asp
UAA	Stop
UAG	Stop
UGC	Cys
UGG	Trp
UUA	Leu
UUG	Leu
UUC	Phe
UUU	Phe

A synthetic mRNA molecule has the following sequence of 30 nucleotides, beginning with a start codon at the 5' end:

AUG UUC CGU GAA UGG CCU GAU UGC UUA GAG

Single-base substitution mutations took place at both the 8th and 14th nucleotide positions in this sequence.

Using **only** the information provided and translating the mRNA sequence from left to right, which of the following statements **could** be correct for the resulting polypeptide?

- 1 This polypeptide could be only four amino acids long.
- 2 The third amino acid in the polypeptide could be unaffected by the mutation.
- 3 If translation does not stop prematurely, the resulting polypeptide could contain seven different amino acids.

- A** none of them
B 1 only
C 2 only

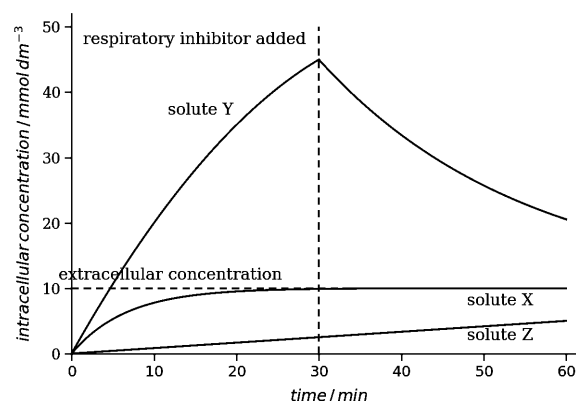
- D 3 only
- E 1 and 2 only
- F 1 and 3 only
- G 2 and 3 only
- H 1, 2 and 3

- 14** Cells of a unicellular eukaryote were placed into a solution containing three different uncharged solutes: X, Y, and Z. The concentration of each solute in the external solution was maintained at a constant 10 mmol dm^{-3} throughout the experiment.

The intracellular concentration of each solute was measured over 60 minutes. At 30 minutes, a respiratory inhibitor was added to the solution. The results are shown in the graph.

Which of the following statements is/are correct?

- 1 The accumulation of solute Y inside the cell between 0 and 30 minutes requires metabolic energy (ATP).
- 2 Solute X is transported into the cell against its concentration gradient.
- 3 After the addition of the respiratory inhibitor at 30 minutes, solute Y leaves the cell by moving down its concentration gradient.



- A none of them
- B 1 only
- C 2 only
- D 3 only
- E 1 and 2 only
- F 1 and 3 only
- G 2 and 3 only
- H 1, 2 and 3

- 15** The allele frequencies in the gene pool of a sexually reproducing population can change over generations as a result of various biological processes and environmental events.

Which of the following would be expected to cause a change in allele frequencies in the gene pool of a population?

1 widespread application of a chemical herbicide across an agricultural weed population

2 immigration of fertile individuals from a genetically distinct neighbouring population

3 a severe population bottleneck caused by a natural disaster that randomly kills the majority of individuals

4 random mating occurring within a very large population in the absence of mutation, migration, and differential selection

A 1 and 2 only

B 1 and 4 only

C 2 and 3 only

D 3 and 4 only

E 1, 2 and 3 only

F 1, 2 and 4 only

G 2, 3 and 4 only

H 1, 2, 3 and 4

- 16** An experiment was conducted to investigate the breakdown of hydrogen peroxide catalysed by catalase extracted from human liver cells.

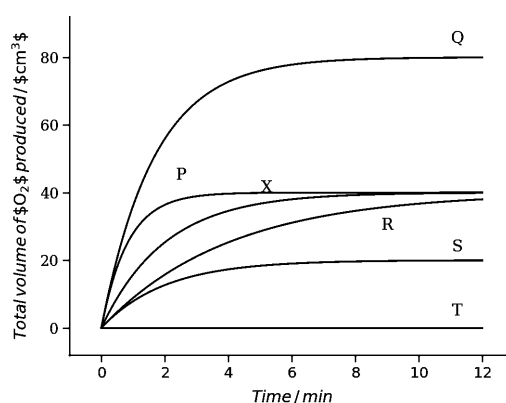
In a control experiment, 10 cm³ of 1.0 mol dm⁻³ H₂O₂ was mixed with 1.0 cm³ of human catalase solution at 25 °C and pH 7.0. The reaction was allowed to proceed to completion, and the total volume of oxygen gas produced was recorded over time, producing curve **X** on the graph.

The experiment was repeated under three separate modifications:

- **Modification 1:** Carried out at 37 °C instead of 25 °C.
- **Modification 2:** Carried out using 10 cm³ of 0.5 mol dm⁻³ H₂O₂ at 25 °C.
- **Modification 3:** Carried out at 25 °C after boiling the catalase solution for 10 minutes and then cooling it.

Which row in the table correctly identifies the curve obtained for each modification?

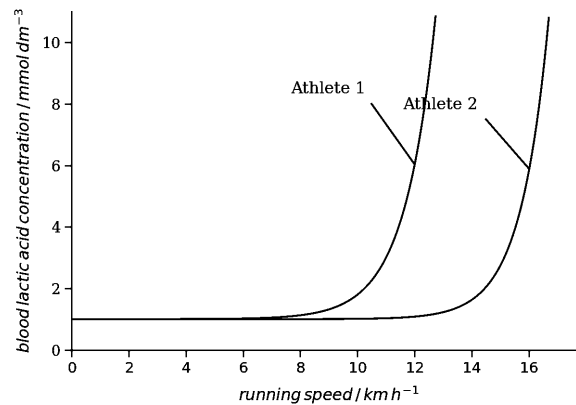
	Modification 1	Modification 2	Modification 3
A	P	S	T
B	P	R	T
C	P	S	R
D	Q	S	T
E	Q	R	T
F	Q	S	R
G	R	S	T
H	R	P	T



- 17 The graph shows the blood lactic acid concentration of two athletes, Athlete 1 and Athlete 2, during an exercise test where running speed was increased incrementally in stages.

Which of the following statements correctly describe(s) the athletes?

- 1 At a running speed of 11 km h^{-1} , Athlete 1 relies more on anaerobic respiration to meet their energy demands than Athlete 2.
- 2 At a running speed of 5 km h^{-1} , the rate of lactic acid production in both athletes is zero.
- 3 If both athletes run at 11 km h^{-1} for 15 minutes, Athlete 1 will incur a larger oxygen debt than Athlete 2.



- A none of them
- B 1 only
- C 2 only
- D 3 only
- E 1 and 2 only
- F 1 and 3 only
- G 2 and 3 only
- H 1, 2 and 3

- 18** Researchers investigated the role of Protein Q in the synthesis of a stress-response metabolite, M. Three bacterial strains were compared: a wild type, a mutant lacking the gene for Protein Q, and a strain engineered to contain high cellular levels of Protein Q. The concentration of metabolite M (arbitrary units) was measured in each strain after incubation at either 20°C or 40°C.

The results are shown in the table below:

Strain	Level of Protein Q	Concentration of M at 20°C	Concentration of M at 40°C
Wild Type	Normal	5 100	5 25
Mutant	None	5 25	5 25
Engineered	High	5 250	5 250

Which of the following conclusions is best supported by the data?

- A** Protein Q is essential for the initiation of metabolite M synthesis.
 - B** Protein Q acts as a temperature sensor that triggers the production of metabolite M.
 - C** The production of metabolite M is proportional to the concentration of Protein Q at both temperatures.
 - D** Protein Q enhances the production of metabolite M but only in the presence of the thermal stimulus.
 - E** The presence of Protein Q suppresses the basal production of metabolite M at 20°C.
 - F** Metabolite M synthesis is regulated solely by the concentration of Protein Q.
- 19** In a species of plant, the allele for tall stems (*T*) is dominant to the allele for dwarf stems (*t*). A tall plant is crossed with a dwarf plant. The resulting offspring show a 1:1 ratio of tall plants to dwarf plants.
- Two tall plants, each with the same genotype as the original tall parent, are then crossed with each other.
- What is the expected proportion of offspring from this second cross that will be dwarf?
- A** 0
 - B** 1/4
 - C** 1/2
 - D** 3/4

- 20** In mice, the allele for black fur, *B*, is dominant to the allele for brown fur, *b*.

Two heterozygous black mice are crossed. A black offspring from this cross is selected at random and crossed with a heterozygous black mouse.

What is the probability that their first offspring has brown fur?

- A** 0
- B** $\frac{1}{12}$
- C** $\frac{1}{6}$
- D** $\frac{1}{4}$
- E** $\frac{1}{3}$
- F** $\frac{1}{2}$

- 21** A diploid cell contains 12 chromosomes. Which row correctly gives the number of chromosomes in each daughter cell after mitosis and in each gamete after meiosis?

- A** Mitosis 6; meiosis 6
- B** Mitosis 6; meiosis 12
- C** Mitosis 12; meiosis 6
- D** Mitosis 12; meiosis 12
- E** Mitosis 24; meiosis 6
- F** Mitosis 24; meiosis 12

- 22** The pedigree shows the inheritance of an X-linked recessive condition in a family.
- Individual 5 has been diagnosed with Klinefelter syndrome (karyotype 47, XXY) and expresses the recessive condition.

Assuming no new mutations occurred and no crossing over took place between the gene locus and the centromere, in which parent did the non-disjunction event occur, and at which stage of cell division?

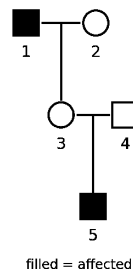


Diagram not to scale

- A** mother; meiosis I
- B** mother; meiosis II
- C** mother; mitosis after fertilisation
- D** father; meiosis I
- E** father; meiosis II
- F** father; mitosis after fertilisation
- G** either parent; meiosis I
- H** either parent; meiosis II
- 23** A culture of secretory cells has a total tissue mass of 10 mg. The average mass of a single cell in the culture is 1 ng.
- Each cell synthesizes a specific protein at a rate of 10^5 molecules per second. The mass of one molecule of this protein is 10^{-18} g.
- What is the total mass of the protein synthesized by the culture in 100 minutes?
- A** 0.1 mg
- B** 1.0 mg
- C** 6.0 mg
- D** 10 mg
- E** 60 mg
- F** 6.0 g

- 24** A suspension contains a mixture of two strains of yeast: a wild-type strain that secretes the enzyme invertase, and a mutant strain that does not. The total concentration of yeast cells in the suspension is 8.0×10^6 cells mL⁻¹ and the invertase activity is 300 arbitrary units (AU) mL⁻¹. A separate control sample containing only the wild-type strain at a concentration of 1.0×10^6 cells mL⁻¹ has an invertase activity of 50 AU mL⁻¹.
- Assuming the mutant strain has zero invertase activity, what is the concentration of the mutant strain in the mixed suspension?
- A** 1.0×10^6 cells mL⁻¹
 - B** 2.0×10^6 cells mL⁻¹
 - C** 4.0×10^6 cells mL⁻¹
 - D** 5.0×10^6 cells mL⁻¹
 - E** 6.0×10^6 cells mL⁻¹

25 The diagram shows a circular plasmid vector containing an origin of replication (*ori*), an ampicillin resistance gene (*Amp^R*), and a β -galactosidase gene (*lacZ*).

The enzyme β -galactosidase hydrolyses the colourless substrate X-gal into a blue product in the presence of the inducer IPTG.

Two recombinant plasmids were produced by inserting a target gene into one of the unique restriction sites:

- **Plasmid A** has the target gene inserted into the *EcoRI* site located within the coding sequence of *lacZ*.
- **Plasmid B** has the target gene inserted into the *PstI* site located within the coding sequence of *Amp^R*.

Samples of ampicillin-sensitive, β -galactosidase-deficient bacteria were separately transformed with either Plasmid A or Plasmid B. The transformed bacteria were spread onto two sets of agar plates:

- **Plate 1:** Nutrient agar containing ampicillin, X-gal, and IPTG
- **Plate 2:** Nutrient agar containing X-gal and IPTG (without ampicillin)

Which row in the table correctly predicts the growth and appearance of each bacterial culture on the two plates?

	Plasmid A on Plate 1 (with ampicillin)	Plasmid A on Plate 2 (without ampicillin)	Plasmid B on Plate 1 (with ampicillin)	Plasmid B on Plate 2 (without ampicillin)
A	blue colonies	blue colonies	white colonies	white colonies
B	white colonies	white colonies	no growth	blue colonies
C	white colonies	blue colonies	no growth	white colonies
D	no growth	white colonies	blue colonies	blue colonies
E	blue colonies	white colonies	no growth	blue colonies
F	white colonies	white colonies	blue colonies	no growth
G	no growth	blue colonies	white colonies	white colonies
H	white colonies	blue colonies	blue colonies	no growth

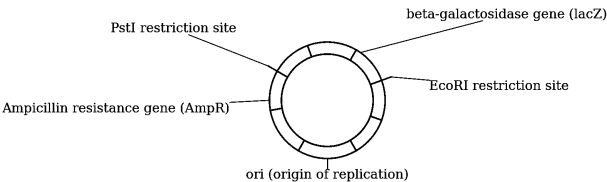


Diagram not to scale

- 26** A functional polypeptide consists of a single chain of 150 amino acids. The complete coding region of the messenger RNA (mRNA) that translates this polypeptide (excluding the stop codon) has the following base composition:

- Adenine (A): 36%
- Uracil (U): 24%
- Cytosine (C): 22%
- Guanine (G): 18%

Which row in the table correctly gives the percentage of thymine (T) in the double-stranded DNA coding region of this gene, and the total number of hydrogen bonds between the complementary base pairs in this DNA region?

	percentage of thymine (T)	total number of hydrogen bonds
A	24%	900
B	24%	1080
C	30%	540
D	30%	1080
E	30%	2160
F	36%	900
G	36%	1080
H	60%	1080

- 27** DNA sequencing was used to analyse a specific gene locus in a child and both biological parents. Paternity and maternity were genetically confirmed. The table below shows the nucleotide base present at a specific position within this locus on both chromosomes for each individual.
- | Individual | Chromosome 1 Base | Chromosome 2 Base |
|------------|-------------------|-------------------|
| Mother | T | T |
| Father | T | T |
| Child | T | A |

Which statement best explains the origin of the Adenine (A) base found in the child?